

# GCP Migration Assessment & Planning Report

Customer Account Platform (APP-001)

## Executive Summary

The Customer Account Platform (APP-001) is a Tier-1, customer-facing stock trading application currently hosted on-premises across two datacenters (DC-East and DC-West). The application processes approximately 15 million transactions annually and has stringent reliability requirements of 99.95% availability, RPO ≤ 5 minutes, and RTO ≤ 60 minutes.

Based on the assessment, the recommended migration approach is Replatform, involving containerization of the existing .NET application and migration of PostgreSQL workloads to managed GCP services.

Given its revenue-critical nature, shared database dependencies, and integration complexity, the application should be migrated during Wave 2 after the Pilot and Wave 1 migrations have validated tooling, deployment patterns, rollback procedures, and operational readiness.

Overall Recommendation: Proceed with migration using a controlled Migration Factory model and Wave 2 execution.

Migration Confidence Score: 87%

## 1. Application Assessment

### Application Profile

| Attribute             | Value                     |
|-----------------------|---------------------------|
| Application Name      | Customer Account Platform |
| Application ID        | APP-001                   |
| Application Type      | .NET                      |
| Business Owner        | Stock Trading             |
| Technical Owner       | Application Team A        |
| Current Hosting       | On-Premises               |
| Primary Data Center   | DC-East                   |
| Secondary Data Center | DC-West                   |
| Customer Facing       | Yes                       |
| Business Criticality  | Tier-1                    |
| Server Count          | 25                        |
| Database Platform     | PostgreSQL                |
| Database Size         | 8 TB                      |

### Business Context

| Metric                    | Value        |
|---------------------------|--------------|
| Annual Transactions       | 15 Million   |
| Peak Transactions Per Day | 120,000      |
| Release Frequency         | Biweekly     |
| Security Classification   | Confidential |
| Compliance Requirements   | PCI, SOX     |

## 2. Business Criticality Assessment

### Classification

Tier-1 Application

The Customer Account Platform directly supports stock trading activities and customer account operations.

### Business Impact of Failure

- Trading disruption
- Customer transaction failures
- Revenue loss
- Regulatory scrutiny
- Customer trust degradation
- Increased operational support costs

Criticality Scorecard

| Area                             | Rating |
|----------------------------------|--------|
| Revenue Impact                   | High   |
| Customer Impact                  | High   |
| Compliance Impact                | High   |
| Operational Impact               | High   |
| Overall Criticality Rating: High |        |

3. Cloud Readiness Assessment

Architecture Readiness

Current state indicates a traditional .NET application architecture requiring containerization.

Assessment

| Area                                | Rating |
|-------------------------------------|--------|
| Application Modernization Readiness | Medium |
| Container Readiness                 | Medium |
| Infrastructure Readiness            | High   |
| Database Readiness                  | High   |
| DevOps Readiness                    | High   |
| Security Readiness                  | High   |
| Operational Readiness               | Medium |

Cloud Readiness Summary

Strengths

- GitHub-based source control
- Existing CI/CD pipeline
- PostgreSQL platform suitability
- Clearly defined HA/DR requirements
- Mature operational ownership

Gaps

- Not yet containerized
- Shared schema dependency
- Tight application coupling
- Complex reporting integrations

Overall Cloud Readiness Score

78 / 100

Assessment: Migration Ready with Moderate Complexity

4. Dependency Assessment

Internal Technical Dependencies

| Dependency              | Criticality |
|-------------------------|-------------|
| Authentication Service  | Critical    |
| Reporting Platform      | Critical    |
| Customer Portal         | Critical    |
| Fraud Detection Service | Critical    |

External Dependencies

| Dependency               | Risk   |
|--------------------------|--------|
| Payment Gateway Provider | Medium |

Data Dependencies

| Dependency      | Risk |
|-----------------|------|
| Shared Database | High |
| Shared Schema   | High |

Dependency Risk Analysis

High-Risk Dependency  
Reporting Platform

Issue:

- Tight coupling exists between application services and reporting components.

Potential Impact:

- Cutover sequencing constraints
- Increased migration testing effort

#### High-Risk Dependency

Shared Database Schema

Issue:

- Shared data model used across connected services.

Potential Impact:

- Database migration coordination required.
  - Additional rollback planning required.
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## 5. 6Rs Migration Assessment

### Migration Strategy Evaluation

| Strategy   | Recommendation       |
|------------|----------------------|
| Rehost     | Not Recommended      |
| Replatform | Strongly Recommended |
| Refactor   | Possible             |
| Repurchase | Not Applicable       |
| Retire     | Not Applicable       |
| Retain     | Not Recommended      |

### Recommended Strategy

Replatform

Justification

The application demonstrates strong suitability for a replatform migration:

- Existing architecture is stable.
- PostgreSQL is fully supported on GCP.
- Containerization is feasible.
- Limited application changes required.
- Lower cost and risk than refactoring.

### Expected Outcome:

- Faster migration
  - Improved resiliency
  - Reduced infrastructure operations overhead
  - Faster deployment cycles
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## 6. Recommended GCP Architecture

### Application Layer

Recommended Platform

Google Kubernetes Engine (GKE)

Benefits:

- High availability
- Horizontal scaling
- Container orchestration
- Rolling deployments
- Improved operational consistency

### Database Layer

Recommended Platform

Cloud SQL for PostgreSQL

Capabilities:

- High availability
- Automated backups
- Point-in-time recovery
- Reduced database administration effort

### Security Layer

Recommended Services:

- IAM
- Secret Manager
- Cloud Audit Logging
- Security Command Center

## Observability Platform

Recommended Services:

- Cloud Monitoring
  - Cloud Logging
  - Alerting Policies
  - SRE Dashboards
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## 7. Migration Wave Recommendation

Recommended Wave

Wave 2

Why Not Pilot?

Pilot wave should be reserved for:

- Low-risk applications
- Lower business impact
- Simpler architectures

APP-001 does not meet those criteria.

Why Wave 2?

The application has:

- Tier-1 business criticality
- Shared database dependencies
- High transaction volume
- External integrations
- Regulatory requirements

Wave 2 provides sufficient time to:

- Validate migration tooling
  - Validate rollback procedures
  - Validate operational runbooks
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## 8. Migration Execution Plan

### Phase 1: Discovery & Assessment

Duration: 2 Weeks

Activities:

- Dependency workshops
- Architecture review
- Database inventory
- Cloud readiness validation

Deliverables:

- Dependency map
- Migration plan
- Risk assessment

### Phase 2: Containerization

Duration: 3 Weeks

Activities:

- Build Docker images
- Security scans
- CI/CD integration

Deliverables:

- Container images
- Deployment templates

### Phase 3: Environment Preparation

Duration: 2 Weeks

Activities:

- GKE provisioned
- Cloud SQL provisioned
- IAM policies configured
- Monitoring enabled

### Phase 4: Database Migration

Duration: 3 Weeks

Activities:

- Replication setup
- Trial migrations
- Data validation

**Phase 5: Testing**

Duration: 3 Weeks

Activities:

- Functional testing
- Regression testing
- Performance testing
- Security testing
- DR testing

**Phase 6: Cutover**

Duration: 1 Week

Activities:

- CAB approval
- Change execution
- Production cutover
- Rollback readiness

**Phase 7: Hypercare**

Duration: 2 Weeks

Activities:

- Production monitoring
- Performance tuning
- Issue remediation
- SRE onboarding

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**9. Reliability Assessment**

**Availability Requirement**

Target: 99.95%

Assessment:

Achievable using GKE and Cloud SQL HA deployment.

**RPO Requirement**

Target: <=5 Minutes

Assessment:

Achievable using managed database replication and backup strategies.

**RTO Requirement**

Target: <= 60 Minutes

Assessment:

Achievable using automated recovery and failover procedures.

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**10. Security Assessment**

**Security Classification**

Confidential

Regulatory Controls Required

PCI

Requirements:

- Encryption
- Access controls
- Audit logging
- Data protection

SOX

Requirements:

- Change management
- Auditability
- Separation of duties

Security Readiness Status

| Area                   | Status   |
|------------------------|----------|
| IAM Controls           | Ready    |
| Audit Logging          | Ready    |
| Secret Management      | Required |
| Vulnerability Scanning | Required |
| Compliance Validation  | Required |

# 11. Testing Strategy

## Functional Testing

- Trading workflows
- Customer account operations
- Authentication validation

## Data Validation

- Record count verification
- Checksum validation
- Data reconciliation

## Performance Testing

- Load testing
- Stress testing
- Latency testing

## Security Testing

- Vulnerability scanning
- IAM review
- Secret validation

## Resiliency Testing

- Zone outage simulation
  - Regional failover testing
  - Disaster recovery exercises
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# 12. Risk Register

| Risk                              | Severity | Mitigation                                    |
|-----------------------------------|----------|-----------------------------------------------|
| Shared Schema Dependency          | High     | Dependency analysis and coordinated migration |
| Tight Reporting Platform Coupling | High     | Refactor integration points before cutover    |
| PostgreSQL Migration Complexity   | High     | Replication-based migration strategy          |
| Vendor Connectivity Failures      | Medium   | End-to-end testing                            |
| Production Cutover Failure        | Medium   | Rollback rehearsal and CAB validation         |

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# 13. Operational Readiness Review

## Monitoring

Status: Required Before Cutover

Requirements:

- Service dashboards
- Infrastructure dashboards
- Capacity monitoring

## Alerting

Required:

- Availability alerts
- Application alerts
- Database alerts
- Security alerts

## Runbooks

Required Before Go-Live:

- Incident Response Runbook
- Database Recovery Runbook
- Deployment Runbook
- Failover Runbook

## SRE Readiness

Assessment: Partially Ready

Required Actions:

- On-call onboarding
  - Operational documentation
  - Escalation validation
  - DR training
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## 14. SRE Handoff Checklist

Prior to ownership transfer:

- Monitoring validated
- Alerting validated
- Runbooks completed
- Escalation procedures approved
- Disaster recovery validated
- On-call readiness confirmed
- Performance baselines established
- Hypercare completed

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## 15. Success Criteria Validation

| Success Metric | Target       | Assessment |
|----------------|--------------|------------|
| Availability   | 99.95%       | Achievable |
| RPO            | ≤ 5 Minutes  | Achievable |
| RTO            | ≤ 60 Minutes | Achievable |
| Data Loss      | Zero         | Achievable |
| Compliance     | PCI/SOX      | Achievable |
| Hypercare Exit | Successful   | Achievable |

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## 16. Executive Recommendation

### Recommendation: PROCEED

The Customer Account Platform is a strong candidate for migration to Google Cloud Platform under the enterprise Migration Factory model.

### Recommended Approach

- Migration Strategy: Replatform
- Target Platform: GKE + Cloud SQL
- Migration Wave: Wave 2
- Migration Complexity: Medium
- Business Risk: High
- Technical Risk: Medium
- Operational Readiness: Requires Additional Preparation

### Leadership Decision

The application should proceed to detailed migration planning, dependency remediation, containerization activities, and Wave 2 scheduling.

### Overall Assessment

| Category                 | Result                 |
|--------------------------|------------------------|
| Cloud Readiness          | 78/100                 |
| Migration Complexity     | Medium                 |
| Business Criticality     | Tier-1                 |
| Recommended Strategy     | Replatform             |
| Recommended Wave         | Wave 2                 |
| Overall Confidence Score | 87%                    |
| Final Recommendation     | Proceed with Migration |